

# Lecture 38

4/27/26

## CONTRAST RECAP

See blackboard notes

Localization & contrast both complicated in MRI

Doctor places patient in strong  $B$  field (3T)  
Protons get biased by field = equilibrium config

This strong field is static  
Come in w/ another  $B$  field that is  
distinct from main field

$B$  fields cause mag to rotate  
Rotate from  $z$  into  $xy$  plane  
This is the signal the doctor measures

$B$  field is a pulse  
Look at  $90^\circ$  pulse (but others possible)

New config is not stable / non equilibrium  
Decay w/ time constants

$M_{xy}$   
 $M_z$

The time constants vary w/ tissue  
These can be basis for generating contrast

Decay & recovery occur over different time scales

They can be a factor of 10 different  
Make contrast depend on these instead of density

Looked at  $T_2$ ,  $T_1$ , &  $\rho$  weighting

|        |                          |                 |
|--------|--------------------------|-----------------|
| $T_2$  | - CSF bright             | Invert contrast |
| $T_1$  | - CSF dark               |                 |
| $\rho$ | - less dramatic contrast |                 |

This ability to control contrast is the strength of MRI

Doctor has two experimentally variable parameters

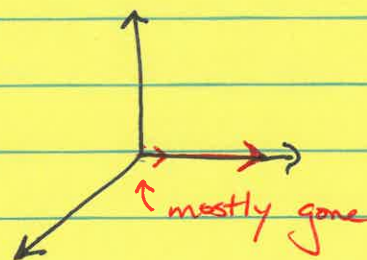
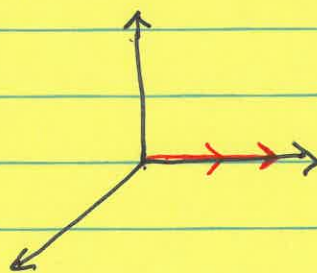
- (1) TR = experiment repetition time  
experiments can be repeated 100s of times
- (2) TE = echo time (when signal is measured)

Conceptual example:

Do  $T_2$  weighting (where  $T_2$  dictates contrast)

$TR = 150 \text{ ms}$  (in this example)

After  
 $90^\circ$  RF pulse  
CSF bright  
White muffled dark



$TE = 150 \text{ ms}$

White matter  $T_2 \approx 70 \text{ ms}$

Skip  $T_1$  argument. It's in the book.

Mathematical Analysis of Contrast

$$SA \propto \rho \left( 1 - e^{-TR/T_1} \right) e^{-TE/T_2}$$

↑ signal amplitude for specific tissue

↑ Returns to equilibrium are exponential

Given  $TR$  &  $TE$

Calculate signal amplitude

Look at arterial blood & white matter

Keep track of every term separately

Ex  $\rho = 0.17$  arterial blood

(1) The bigger SA is the brighter tissue

(2) Whatever term is most different determines weighting protocol

$$\rho (1 - e^{-TR/T_1}) e^{-TE/T_2}$$

|     |     |     |
|-----|-----|-----|
| 0.7 | 0.5 | 0.3 |
| 0.6 | 0.1 | 0.4 |

↑ biggest difference  $\Rightarrow$  T1 weighting

### ACTIVITY 14

$$SA_{AB} = \rho (1 - e^{-TR/T_1}) e^{-TE/T_2}$$

$$(0.72) (1 - e^{-2000/1932}) e^{-70/275}$$

$$0.72 \cdot 0.64 \cdot 0.78 = 0.36$$

$$SA_{WM} = 0.61 (1 - e^{-2000/1080}) e^{-70/70} = 0.19$$

$$0.61 \cdot 0.84 \cdot 0.37$$

two-fold difference

MRI Strengths & Weaknesses See list on lower 10 int. slides  
 Looked at exemplary Pulse Sequence

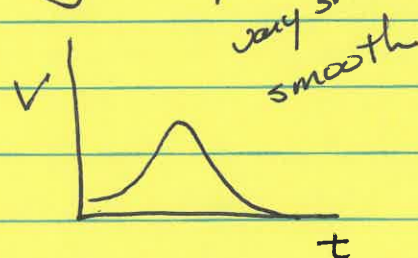
Strengths  
 Low  $\gamma$  Radiation  
 Sub-mm high-contrast resolution of soft tissue  
 Functional MRI available

Weaknesses  
 Heating  
 Not compatible w/ metal  
 Expensive!

## DIGITAL IMAGE PROCESSING

Adjust parameters to improve image quality  
 Analyze image data

Compare digital & analog signals



Sample analog signals to generate digital signal. Need to be aware of noise & sampling theorem

## Key steps in <sup>acquiring</sup> digital images

① Sampling - must satisfy sampling theorem

e.g., want to achieve <sup>diffraction-</sup> limited resolution of 200 nm you must sample at  $< 100$  nm



This is true even if optics is good

Camera pixels typically set at  $\sim 100$  nm for this reason

② Quantization of signal strength

Camera has a certain number of gray levels to encode brightness

Ex 8-bit detector  $2^8 = 256$

$\Rightarrow 0 - 255$

This is a property of detector  $\uparrow$  black  $\uparrow$  white

Assign a value that is quantized in jumps

Price increases as b.t number increases



One established <sup>restoration</sup> protocol

- 1 Flat field correction - remove gradients in illumination
- 2 Brightness & Contrast restoration e.g. bad folder
- 3  $\gamma$  correction - bring up dim features
- 4 Noise reduction
- 5 Fine detail restoration

Know  
these 3